EXPERIMENT-18:

AIM:

To write a C program to demonstrate DES subkey generation, where the 56-bit key is divided into two 28-bit halves, and each round subkey is generated by shifting and selecting bits from both halves.

CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
   char key[57], C[29], D[29];
   int i, shift[] = {1,1,2,2,2,2,2,2,1,2,2,2,2,2,2,1};

   printf("Enter 56-bit key: ");
   scanf("%56s", key);

   for(i=0;i<28;i++) C[i]=key[i];
   C[28]='\0';

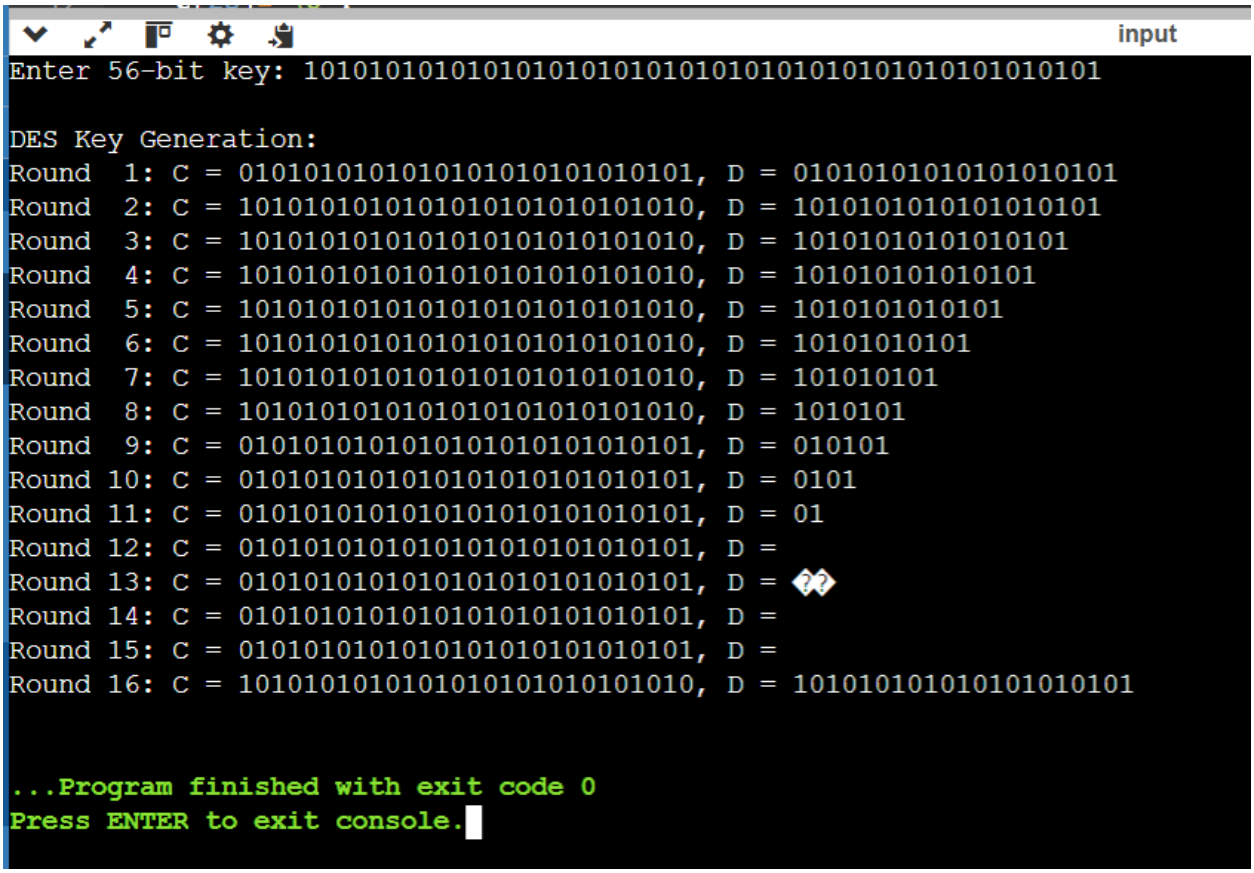
   for(i=0;i<28;i++) D[i]=key[i+28];
   D[28]='\0';

   printf("\nDES Key Generation:\n");
   for(i=0;i<16;i++) {
      int s=shift[i];
```

```
        while(s--) {
            char a=C[0], b=D[0];
            memmove(C,C+1,27); C[27]=a;
            memmove(D,D+1,27); D[27]=b;
        }
        printf("Round %2d: C = %.28s, D = %.28s\n",
              i+1,C,D);
    }

    return 0;
}
```

OUTPUT:


input
Enter 56-bit key: 1010101010101010101010101010101010101010101010101010101
DES Key Generation:
Round 1: C = 0101010101010101010101010101, D = 01010101010101010101
Round 2: C = 1010101010101010101010101010, D = 1010101010101010101
Round 3: C = 1010101010101010101010101010, D = 10101010101010101
Round 4: C = 1010101010101010101010101010, D = 101010101010101
Round 5: C = 1010101010101010101010101010, D = 1010101010101
Round 6: C = 1010101010101010101010101010, D = 10101010101
Round 7: C = 1010101010101010101010101010, D = 101010101
Round 8: C = 1010101010101010101010101010, D = 1010101
Round 9: C = 0101010101010101010101010101, D = 010101
Round 10: C = 0101010101010101010101010101, D = 0101
Round 11: C = 0101010101010101010101010101, D = 01
Round 12: C = 0101010101010101010101010101, D =
Round 13: C = 0101010101010101010101010101, D =
Round 14: C = 0101010101010101010101010101, D =
Round 15: C = 0101010101010101010101010101, D =
Round 16: C = 1010101010101010101010101010, D = 10101010101010101010
...Program finished with exit code 0
Press ENTER to exit console.